

Deepak Mallampati

APPLIED AI / MLOPS ENGINEER

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PROFESSIONAL SUMMARY

MLOps and Applied AI Engineer with 4+ years keeping production ML and LLM systems reliable at scale. Own the infrastructure spanning Linux, Docker, and Kubernetes up through the ML stack - model deployment, monitoring, and CI/CD - turning research artifacts into robust, continuously improving services. Cut per-query cost and latency (p95 1.5s to 800ms) while sustaining SLA targets across 100,000+ daily requests, reduced failed production runs by 60% with fault-tolerant orchestration, and built LLM evaluation and observability frameworks (RAGAS, BERTScore, latency/accuracy dashboards) that surface regressions before release.

CORE COMPETENCIES

MLOps Lifecycle • Kubernetes & Docker • CI/CD (Jenkins, Azure DevOps) • Model Deployment & Serving • Monitoring & Observability • GCP / AWS / Azure • Model Optimization & Throughput • MLflow / Weights & Biases • Python / Scientific Stack

MLOps & Infrastructure: Docker, Kubernetes, Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model Monitoring & Observability | Drift Detection | Semantic Caching | Model Routing

Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services), GCP Vertex AI

ML & LLMs: PyTorch, TensorFlow, Keras | RAG Architecture | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | RAGAS, BERTScore Evaluation | Hugging Face Transformers

Data & Serving: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | FastAPI | FAISS, Pinecone, Weaviate

Languages: Python, JavaScript/TypeScript, Java, SQL

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer - USA

- Optimized inference **cost and latency** through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead across production LLM services.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large-scale healthcare corpora, combining dense retrieval with cross-encoder re-ranking to lift relevance and reduce hallucination.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern - USA

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across 25+ AI agents.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by 27% on internal evaluation datasets.
- Delivered scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams, partnering with product, platform, and data engineering.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant - USA

- Built a **privacy-preserving ML pipeline** (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from 3.38% to 0.32% while retaining 99% of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset for controllable long-form generation.

Emerson

Jun 2022 - Apr 2023

ML Trainee - India

- Developed and trained supervised **regression and classification models** on operational data for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving 89% F1 on custom NER tasks.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - production deployment, monitoring, and iteration on live user traffic.

career-ops

Autonomous AI-driven job-search pipeline that scans, scores, and generates tailored applications overnight - scheduled orchestration with retry logic and reliability guarantees.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data, quantifying the privacy-utility trade-off across configurations.

EDUCATION

M.S. in Computer Science, Kent State University, OH

2025 · GPA 3.70/4.0